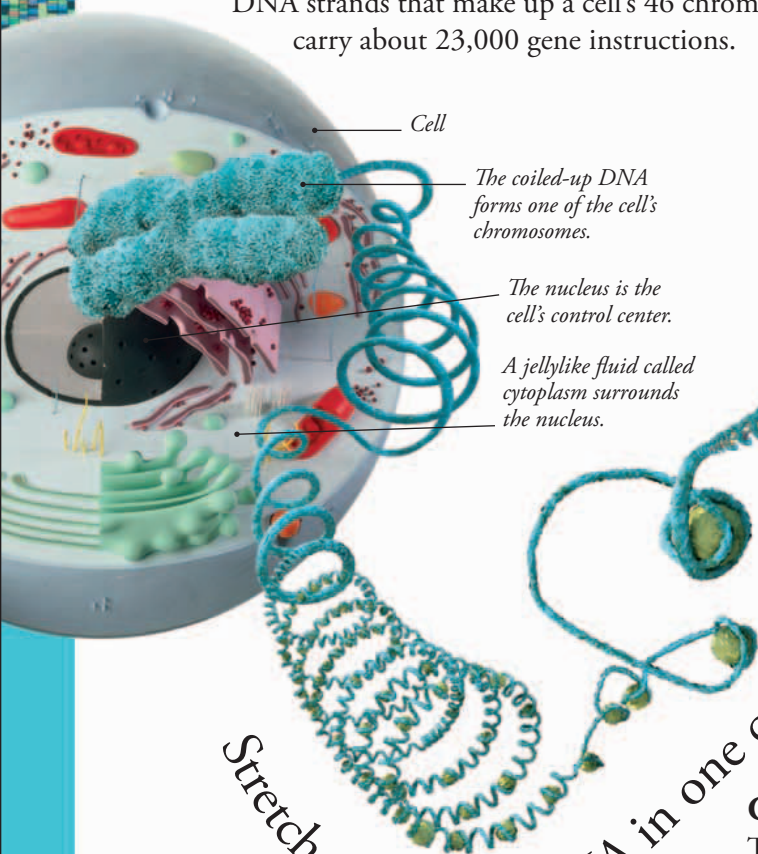


Code of life

Inside every one of your cells is a unique set of instructions needed to build and maintain your body. Except for identical twins, no one in the world has exactly the same set of instructions. This genetic information is carried by a substance called DNA found inside the nucleus of each cell.

SUPER SPIRAL

Within a cell's nucleus are X-shaped structures called chromosomes. Each chromosome is made of tightly coiled-up DNA. Unraveled, DNA can be seen to be made of two long, linked strands that spiral around each other, resembling a twisted ladder. Together the DNA strands that make up a cell's 46 chromosomes carry about 23,000 gene instructions.

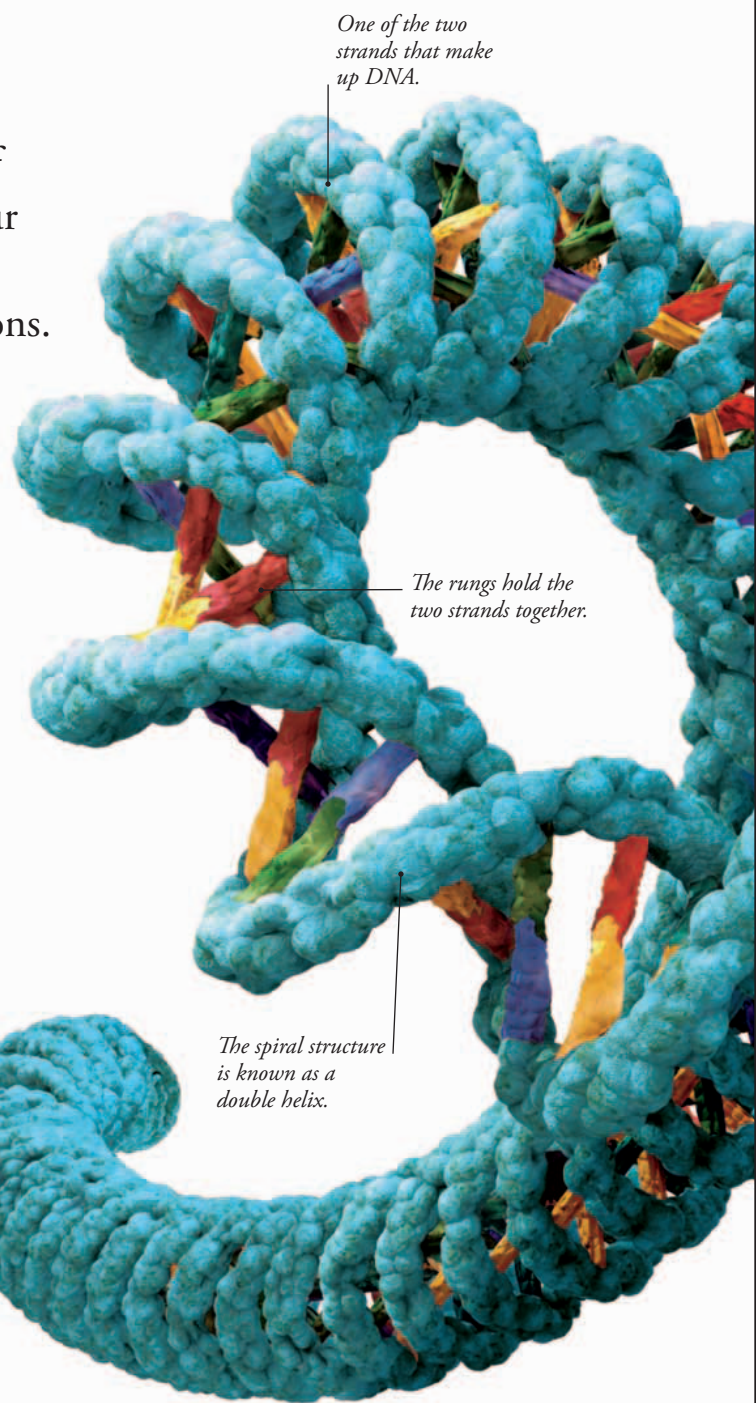


Cell

The coiled-up DNA forms one of the cell's chromosomes.

The nucleus is the cell's control center.

A jellylike fluid called cytoplasm surrounds the nucleus.



One of the two strands that make up DNA.

The rungs hold the two strands together.

The spiral structure is known as a double helix.

Stretched out, the DNA in one cell alone would reach 6 ft (1.8 m).

CODED MESSAGES

The rungs of the DNA ladder consist of four chemicals called bases. They form the coded instructions for making proteins. These vital substances are essential for making sure that your body works—your hair, skin, and muscles are all made from proteins.

The four bases

- Guanine
- Cytosine
- Thymine
- Adenine